Example 8.5

Calculate the pH of 0.1M CH_3COOH solution. Dissociation constant of acetic acid is 1.8×10^{-5}.

$pH=-\log[H^+]$

For weak acids,

$[H^+]=\sqrt{K_a\times C}$

$=\sqrt{1.8\times10^{-5}\times0.1}$

$=1.34\times10^{-3}M$ $\qquad pH=-\log\left(1.34\times10^{-3}\right)$

$=3-\log1.34$

$=3-0.1271$

$=2.8729\simeq2.87$

Evaluate yourself - 7

K_b for NH_4OH is 1.8×10^{-5}. Calculate the percentage of ionisation of 0.06M ammonium hydroxide solution.

8.6 Common Ion Effect

When a salt of a weak acid is added to the acid itself, the dissociation of the weak acid is suppressed further. For example, the addition of sodium acetate to acetic acid solution leads to the suppression in the dissociation of acetic acid which is already weakly dissociated. In this case, CH_3COOH and CH_3COONa have the common ion, CH_3COO^-

Let us analyse why this happens. Acetic acid is a weak acid. It is not completely dissociated in aqueous solution and hence the following equilibrium exists.

$$CH_3COOH(aq)\rightleftharpoons H^+(aq)+CH_3COO^-(aq)$$

However, the added salt, sodium acetate, completely dissociates to produce Na^+ and CH_3COO^- ion.

$$CH_3COONa(aq)\rightarrow Na^+(aq)+CH_3COO^-(aq)$$

Hence, the overall concentration of CH_3COO^- is increased, and the acid dissociation equilibrium is disturbed. We know from Le chatelier's principle that when a stress is applied to a system at equilibrium, the system adjusts itself to nullify the effect produced by that stress. So, inorder to maintain the equilibrium, the excess CH_3COO^- ions combines with H^+ ions to produce much more unionized CH_3COOH i.e, the equilibrium will shift towards the left. In other words, the dissociation of CH_3COOH is suppressed. Thus, the dissociation of a weak acid (CH_3COOH) is suppressed in the presence of a salt (CH_3COONa) containing an ion common to the weak electrolyte. It is called the common ion effect.

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